

Large-Scale and Multi-Structured Databases

Project Design

Distributed Book Social Media

Danilo Parisi

Matteo Iori

Matteo Billi Ciani

Application Highlights

Distributed Book Social Media

Book Discovery

- Browse trending and recommended books
- Search by title, author, genre etc.
- Find similar books
- See reviews and books' popularity

Users' Experience

- Add books to the “*read list*” and the “*want-to-read*” list
- Submit books reviews and their votes
- Follow other users and get their influence
- Get suggestions for similar liked books

Administrative Operations

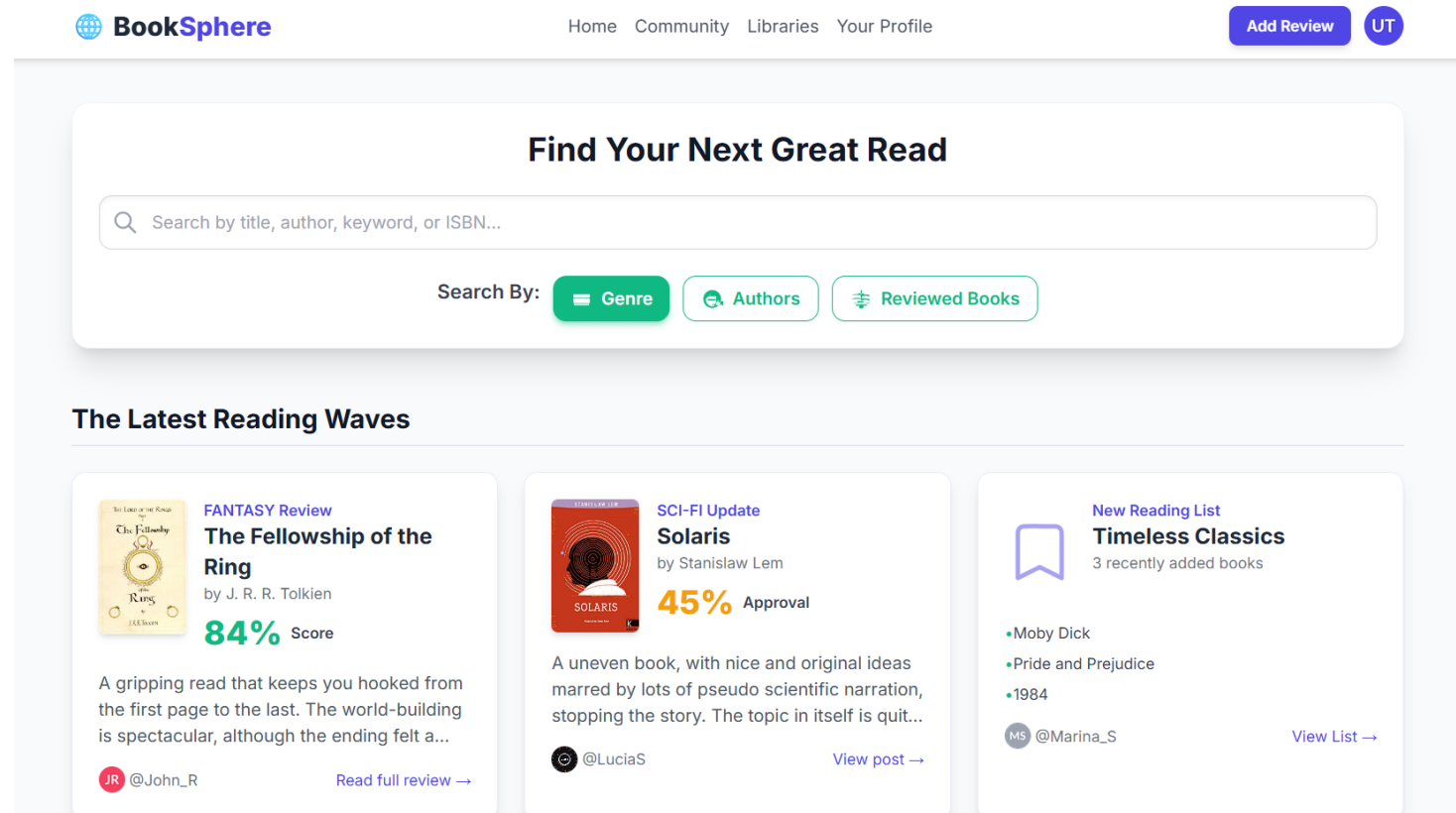
- Moderate Users' behaviors (delete reviews, block users etc.)
- Add new books, authors and genres to the catalogue
- Manage analytics over the system (ranking of the year, index of internationality, popularity predictions for newly added books etc.)

Actors and main mock-ups (i)

Actors:

- Unregistered User
- Standard User
- Admin

User Browsing Page



Actors and main mock-ups (ii)

Admin Page

BookSphere
Review Moderation

Recent Reviews

Analytics

User Management

Review Moderation Dashboard

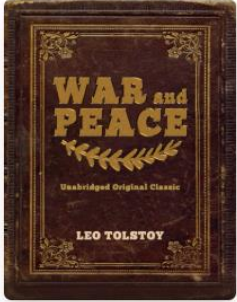
BOOK TITLE & AUTHOR	USER	SCORE	DATE	ACTIONS
The Last Dragon by Jane Doe	@Alice_R	92%	1 hour ago	<button>Advise User</button> <button>Delete</button>
City of Shadows by Marc Brown	@Bob_S	55%	3 hours ago	<button>Advise User</button> <button>Delete</button>
The Bland Cookbook by Anonymous	@Charlie_V	28%	5 hours ago	<button>Advise User</button> <button>Delete</button>
Symphony of Stars by Alex Turner	@Space_Lvr	78%	6 hours ago	<button>Advise User</button> <button>Delete</button>
The Quiet Historian by Lisa Klein	@Hist_Fanatic	68%	7 hours ago	<button>Advise User</button> <button>Delete</button>
My Dog Ate It by John Smith	@JSmith88	35%	8 hours ago	<button>Advise User</button> <button>Delete</button>
The Martian Colony by Maria Lee	@SciFi_Guy	88%	9 hours ago	<button>Advise User</button> <button>Delete</button>
Romantic Paris by Jean Pierre	@DateNight	48%	10 hours ago	<button>Advise User</button> <button>Delete</button>
The CEO's Guide by Business Guru	@BigMoney	72%	11 hours ago	<button>Advise User</button> <button>Delete</button>

Actors and main mock-ups (iii)

Book Discovery

 **Booksphere**

War and Peace



My Notes / Summary:

War and Peace is an epic novel tracing the history of the French invasion of Russia and the impact of the Napoleonic era on Tsarist society through the stories of five aristocratic Russian families. It is a masterpiece that explores themes of history, war, family, and philosophy.

Book Details

Author	Leo Tolstoy
Genre	Historical Fiction / Classic
Published	1869
Pages	~1,225 (varies)
ISBN	978-0140447939 (Example)

Average Community Score: **78%**

Internationality Index Visualization

This chart visualizes the distribution of reading activity across different regions and cohorts.



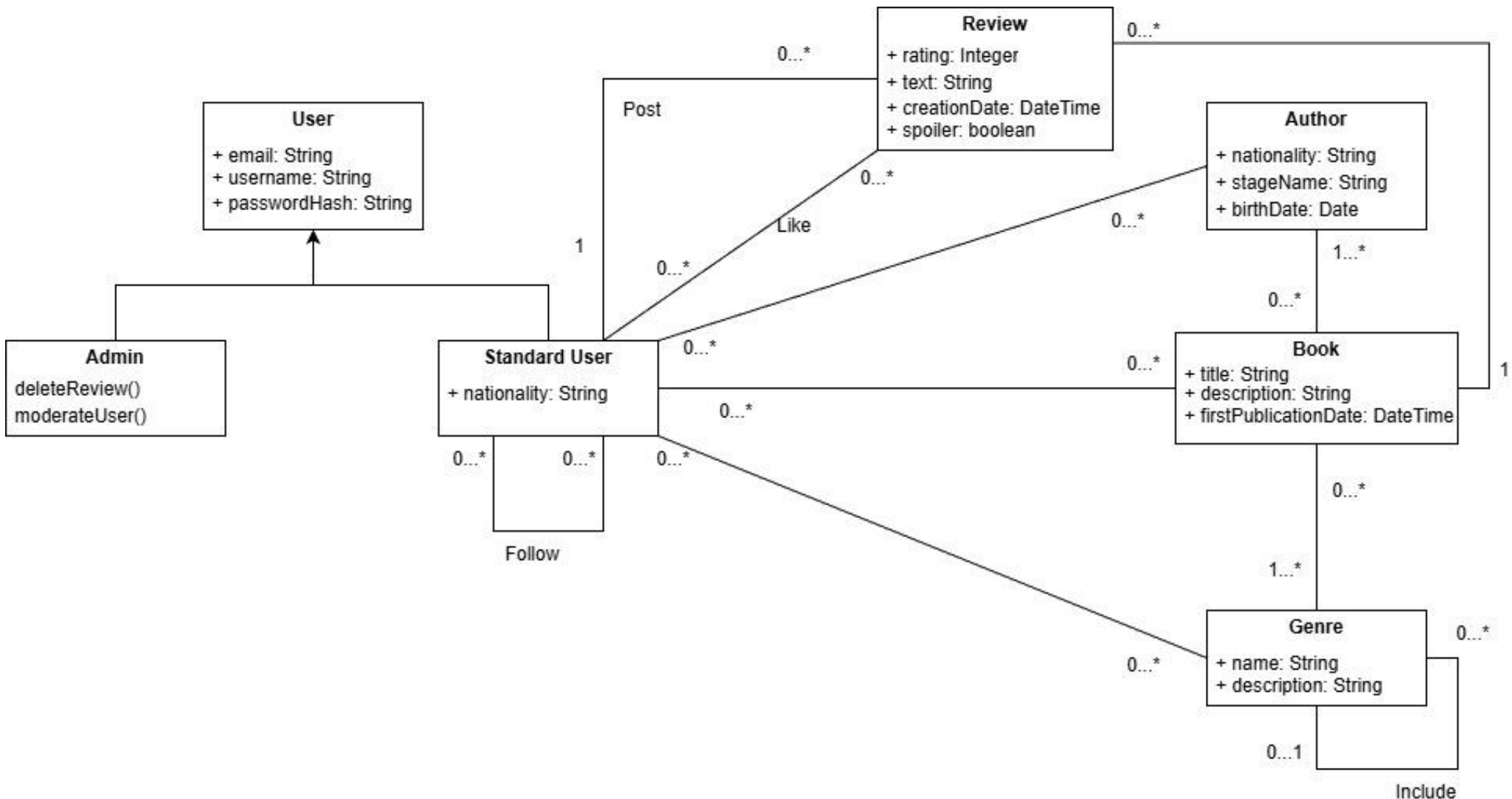
Dataset Description

<i>Source</i>	<i>Description</i>	<i>Volume</i>
<u>goodbooks10k</u>	Contains six million user ratings for the 10,000 books. It also includes books metadatas (title, author, etc.).	~ 90MB
<u>Randomuser API</u>	Dynamic creation of fake users and reviews. This API allows to generate fully fictional user data on demand.	Dynamic creation of fake users and reviews
<u>Amazon Reviews dataset</u>	Amazon review Dataset contains product reviews and metadata from Amazon, spanning May 1996 - July 2014.	~ 5GB

Variety: the data are collected from different sources such APIs, real reviews from Amazon, and randomly generated user data

Velocity/Variability: Old reviews (1 year) are not meaningful for some operations based on a specific period, such as the *end year recap*

Preliminary UML Class Diagram



Relevant Queries and Entities handled by Document DB

List of most relevant functional requirements supported by the Document DB

- The system must calculate the **average ratings** for authors and books;
- The system must **find highest ranked books** by genre/author/year;
- The system must compute the **most read books** for a specified year;
- The system must allow Admin to see a **popularity prediction** for a new publication;
- The system must perform an **end year recap** for the user: most liked genres, authors and new discoveries.

List of entities stored in the Document DB

- Book
- Author
- Genre
- Review
- User

Requirements and Entities handled by Graph DB

List of the most relevant functional requirements supported by the Graph DB

- The system must **identify influencers** inside the application;
- The system must compute an **author versatility** in term of covered genres;
- The system must compute an **internationality index** of an author/book based on where it has been liked/read and how far it has gone;
- The system must permit to **generate recommendations** for a Standard User.

List of the entities stored in the Graph DB

- User
- Book
- Author
- Genre

